THE RADIUS CRITERION IN REAL ANALYTIC AND LOWER REGULARITY: AN EXPANDED PROOF AND AUDIT

Abstract. For a holomorphic line bundle A over a compact complex manifold, the total space of A^{-1} admits a real analytic plurisubharmonic exhaustion if and only if A admits a real analytic Hermitian metric with semipositive curvature. We prove the equivalence, together with its continuous, differentiable, and local Hölder versions. The proof includes the uniform analyticity estimate needed for circle averaging, a derivative-free positivity argument for the logarithmic radius, and a separate verification of every regularity class. We also correct one inference about regularity inclusions in the preceding discussion.

Contents

1. Statement, conventions, and scope 1
2. Circle averaging, including real analyticity 2
3. Logarithmic radii from a continuous exhaustion 4
3.1. Convexity without classical second derivatives 5
3.2. Choice and continuity of the radius 5
4. Positivity of the negative logarithmic radius 6
4.1. Gluing and the sign of the curvature 7
5. Recovering precisely the original regularity 7
5.1. Classical C^r and real analytic regularity 7
5.2. The $C^{0,\alpha}$ case, including Lipschitz regularity 7
5.3. The $C^{k,\alpha}$ case for $k \geq 1$ 8
6. From a semipositive metric to an exhaustion 8
6.1. Regularity, including the zero section 9
6.2. Plurisubharmonicity with no loss at zero 9
6.3. Compactness of the sublevels 9
7. The classical Levi calculations, written out 10
7.1. Levi form of the radius 10
7.2. Levi form of the squared dual norm 10
8. Audit of the earlier discussion 11
References 12

1. Statement, conventions, and scope

The answer to the real analytic question is *yes*. The relevant equivalence is at the *same* regularity on both sides; it does not assert that a smooth exhaustion or metric can be replaced by a real analytic one. The following statement extends Proposition 3.3 and Corollary 3.4 of the manuscript [1] and incorporates the low-regularity argument discussed in the chat entitled *"Generalizing Corollary 3.4"* (the title translated from Chinese).

Date: November 2, 2026.

Theorem 1.1 (Radius criterion at each regularity)**.** *Let Z be a compact complex manifold, let $A \to Z$ be a holomorphic line bundle, and put $M = \mathrm{Tot}(A^{-1})$. Choose any one of the classes*

$$\mathcal{R} \in \{C^r : r \in \mathbb{Z}_{\geq 0} \cup \{\infty\}\} \ \cup \ \{C^{k,\alpha}_{\mathrm{loc}} : k \in \mathbb{Z}_{\geq 0},\ 0 < \alpha \leq 1\} \ \cup \ \{C^\omega\}. \tag{1.1}$$

Then the following are equivalent:

(i) *M admits a real-valued plurisubharmonic exhaustion of class $\mathcal{R}$.*
(ii) *A admits a positive definite Hermitian metric of class $\mathcal{R}$ whose local weights are plurisubharmonic.*

In particular,

$$\begin{gathered} \mathrm{Tot}(A^{-1}) \text{ admits a real analytic psh exhaustion} \\ \Longleftrightarrow \\ A \text{ admits a real analytic semipositive Hermitian metric.} \end{gathered} \tag{1.2}$$

No Kähler or projectivity hypothesis on Z is needed.

An *exhaustion* here is a continuous real-valued function F such that $\{F \leq a\}$ is compact for every $a \in \mathbb{R}$. We use

$$d^c = \frac{i}{4\pi}(\bar\partial - \partial), \qquad dd^c = \frac{i}{2\pi}\partial\bar\partial.$$

For a holomorphic frame e_i of A over $B_i \subset Z$, write

$$\|e_i\|_h^2 = h_i = e^{-\psi_i} > 0. \tag{1.3}$$

The metric is *semipositive* if $\psi_i \in \mathrm{PSH}(B_i)$. Below C^2 regularity this means $dd^c\psi_i \geq 0$ as a current; it does not require pointwise second derivatives. The positivity of h_i in (1.3) refers to the fiber norm, whereas semipositivity refers to its curvature.

All regularity is local in real coordinates underlying holomorphic charts. For $0 < \alpha \leq 1$, $f \in C^{k,\alpha}_{\mathrm{loc}}(\Omega)$ means $f \in C^k(\Omega)$ and, for each compact $K \Subset \Omega$,

$$|D^\beta f(x) - D^\beta f(y)| \leq C_K |x - y|^\alpha \quad (x, y \in K,\ |\beta| = k).$$

In particular, $C^{0,1}_{\mathrm{loc}}$ means locally Lipschitz. At integer orders C^r has its classical meaning; $C^{k,1}_{\mathrm{loc}}$ is not identified with C^{k+1}. If a real noninteger index is desired, our convention is $C^{k+\alpha} = C^{k,\alpha}_{\mathrm{loc}}$ for $0 < \alpha < 1$. Real analyticity, denoted C^ω, means representability by a convergent power series in the real coordinates near every point. No single analytic radius or Hölder constant on all of M is required.

The metric coefficients h_i, their reciprocals h_i^{-1}, and the weights $-\log h_i$ have the same membership in each of the classes in (1.1). Indeed, on every sufficiently small relatively compact chart, h_i is bounded away from zero, and composition with log, exp, or $s \mapsto 1/s$ preserves the indicated regularity. For the analytic class these functions are themselves real analytic on their domains.

2. Circle averaging, including real analyticity

Let $R_\theta(v) = e^{i\theta}v$ be the fiberwise circle action and let F_0 be an exhaustion as in Theorem 1.1(i). Define

$$F(v) = \frac{1}{2\pi}\int_0^{2\pi} F_0(R_\theta v)\, d\theta. \tag{2.1}$$

Lemma 2.1. *The function F is a circle-invariant psh exhaustion. It belongs to every class in (1.1) to which F_0 belongs.*

Proof of continuity, invariance, positivity, and properness. For a compact set $K \subset M$, the orbit set $S^1 \cdot K$ is compact. The continuous integrand in (2.1) is uniformly continuous on compact coordinate neighborhoods times S^1; hence its integral is continuous. Changing the variable θ shows $F(R_\eta v) = F(v)$.

Every R_θ is holomorphic. For a holomorphic disc f defined on a neighborhood of $\overline{\Delta}$ in M, the submean inequality for $F_0 \circ R_\theta \circ f$, integrated in θ, gives

$$F(f(0)) \le \frac{1}{2\pi} \int_0^{2\pi} F(f(e^{is}))\, ds.$$

The interchange of integrals is valid because the integrand is continuous on a compact set. This proves that F is psh.

Put $K_a = \{F_0 \le a\}$. If $F(v) \le a$, at least one point of its circle orbit lies in K_a: otherwise the continuous function $\theta \mapsto F_0(R_\theta v)$ would have minimum strictly larger than a, contrary to its average. Therefore

$$\{F \le a\} \subset S^1 \cdot K_a. \tag{2.2}$$

The right side is compact, being the continuous image of $S^1 \times K_a$. The left side is closed, so it is compact as well. Thus F is an exhaustion. □

Proof of finite, smooth, and Hölder regularity. In a bundle chart use real coordinates $x = (z, \lambda)$, with complex coordinates regarded as pairs of real coordinates. Rotation is the linear map $T_\theta(z, \lambda) = (z, e^{i\theta}\lambda)$, whose matrix entries and operator norm are uniformly bounded in θ. For $F_0 \in C^r$ and $|\beta| \le r$,

$$D^\beta F(x) = \frac{1}{2\pi} \int_0^{2\pi} D_x^\beta (F_0 \circ T_\theta)(x)\, d\theta. \tag{2.3}$$

The derivatives of the integrand are jointly continuous and uniformly bounded on the relevant compact sets. Difference quotients and dominated convergence justify (2.3). Its right side is continuous. Applying this for every finite r proves the C^∞ assertion as well.

For $F_0 \in C^{k,\alpha}_{\mathrm{loc}}$, the chain rule for the linear maps T_θ and the local Hölder estimates on the compact orbit set give, on a smaller compact coordinate neighborhood,

$$\left|D^\beta (F_0 \circ T_\theta)(x) - D^\beta (F_0 \circ T_\theta)(y)\right| \le C|x - y|^\alpha, \qquad |\beta| = k,$$

with C independent of θ. A finite neighborhood cover of the orbit set supplies such a uniform constant; pairs of points not close enough to lie in one of these neighborhoods are handled by boundedness of the derivatives. Integrating the estimate proves the same Hölder bound for $D^\beta F$. This also applies when $k = 0$ or $\alpha = 1$. □

Lemma 2.2 (Analytic integration over a compact parameter)**.** *Suppose $G(x, \theta)$ is jointly real analytic on a neighborhood of $\{x_0\} \times S^1$, where $x \in \mathbb{R}^m$. Then*

$$I(x) = \frac{1}{2\pi} \int_0^{2\pi} G(x, \theta)\, d\theta$$

is real analytic near x_0.

Proof. Use a real angular coordinate near each point of S^1. The convergent real power series of G in (x, θ) extends holomorphically to the complexified variables on a smaller polydisc. After shrinking that polydisc, its extension is bounded there. Consequently, for each angular point there is an arc J, a number $R_J > 0$, and a bound M_J such that, for every real $\theta \in J$, the function $x \mapsto G(x, \theta)$ has a bounded holomorphic extension to

$$|x_j - x_{0,j}| < 2R_J, \qquad 1 \le j \le m.$$

Compactness of S^1 supplies finitely many such arcs. Taking a smaller common radius and the largest of the finitely many bounds, Cauchy's estimates give constants $R, M > 0$ such that

$$\left| \frac{D_x^\beta G(x_0, \theta)}{\beta!} \right| \leq M R^{-|\beta|} \quad \text{for every multi-index } \beta \text{ and every } \theta. \tag{2.4}$$

Each local extension has the given real function as its restriction, so its Taylor coefficients in x are precisely those in (2.4). No choice of a global complexification in the angular variable is needed.

For $|x - x_0|_\infty < R$, the Taylor series

$$G(x, \theta) = \sum_{\beta \in \mathbb{Z}^m_{\geq 0}} \frac{D_x^\beta G(x_0, \theta)}{\beta!} (x - x_0)^\beta$$

converges uniformly in θ on every smaller polydisc. Indeed, if $|x - x_0|_\infty \leq r < R$, the sum of the absolute bounds is at most

$$M \sum_\beta (r/R)^{|\beta|} = M(1 - r/R)^{-m} < \infty.$$

Termwise integration is therefore justified and yields the convergent power series

$$I(x) = \sum_\beta \left(\frac{1}{2\pi} \int_0^{2\pi} \frac{D_x^\beta G(x_0, \theta)}{\beta!} \, d\theta \right) (x - x_0)^\beta.$$

Thus I is real analytic near x_0. □

Analytic case of Lemma 2.1. In a bundle chart the function

$$G(x, \theta) = F_0(T_\theta x)$$

is jointly real analytic: the rotation matrix has analytic entries $\sin\theta$ and $\cos\theta$, and F_0 is real analytic. Apply Lemma 2.2 at each x_0. This proves $F \in C^\omega$, including at the zero section. □

Remark 2.3. Differentiating the integral to every order proves smoothness, but does not by itself prove analyticity. The uniform convergence radius, or equivalently the factorial estimates (2.4), is the additional ingredient. There is also no appeal to approximation of smooth psh functions by analytic ones.

3. Logarithmic radii from a continuous exhaustion

Fix a holomorphic frame e_i of A on a coordinate neighborhood B_i. Write vectors of A^{-1} as $v = \lambda_i e_i^*(z)$ and set

$$U_i(z, t) = F\big(e^{t/2} e_i^*(z)\big), \qquad V_i(z, w) = F\big(e^{w/2} e_i^*(z)\big). \tag{3.1}$$

The map $(z, w) \mapsto e^{w/2} e_i^*(z)$ is holomorphic. Consequently V_i is psh on $B_i \times \mathbb{C}$, and circle invariance gives

$$V_i(z, w) = U_i(z, \operatorname{Re} w). \tag{3.2}$$

3.1. **Convexity without classical second derivatives.** For fixed z, write $u(t) = U_i(z,t)$ and $w = t + is$. Subharmonicity of $u(\operatorname{Re} w)$ implies, as distributions,

$$\Delta_w(u(t)) = u''(t) \geq 0.$$

To read this as an inequality on the real line, test against a product $\eta(t)\chi(s)$ with $\eta, \chi \geq 0$ and $\int \chi = 1$; the term involving χ'' integrates to zero. Thus $\langle u'', \eta \rangle \geq 0$. Convolve u on compact subintervals with a nonnegative smooth approximate identity. The resulting smooth functions satisfy $u_\varepsilon'' \geq 0$ and are convex. They converge locally uniformly to the continuous function u. Passing to the limit in the convexity inequality proves that u is convex.

Moreover,

$$\lim_{t \to -\infty} U_i(z,t) = F(0_z), \qquad \lim_{t \to +\infty} U_i(z,t) = +\infty. \tag{3.3}$$

The first limit is continuity at the zero vector. If the second failed, there would be a sequence $t_\nu \to +\infty$ along which $U_i(z, t_\nu)$ is bounded above. Its corresponding vectors would lie in one compact sublevel of F, although their fiber coordinates tend to infinity. This is impossible.

Lemma 3.1 (Unique upper-level radius)**.** *Let $u : \mathbb{R} \to \mathbb{R}$ be continuous and convex, with $\lim_{t \to -\infty} u(t) = \ell \in \mathbb{R}$ and $\lim_{t \to +\infty} u(t) = +\infty$. Then u is nondecreasing. For every $c > \ell$ there is a unique $s \in \mathbb{R}$ such that $u(s) = c$, and*

$$u(t) < c \iff t < s, \qquad u(t) \leq c \iff t \leq s.$$

If u is differentiable at s, then $u'(s) > 0$.

Proof. If $u(b) < u(a)$ for some $a < b$, let $d = (u(b) - u(a))/(b - a) < 0$. The monotonicity of secant slopes for convex functions implies, for $x < a$,

$$\frac{u(a) - u(x)}{a - x} \leq d, \qquad u(x) \geq u(a) - d(a - x) \longrightarrow +\infty \quad (x \to -\infty),$$

contradicting the finite limit. Hence u is nondecreasing, and $u \geq \ell$. The intermediate value theorem supplies a point at level c. If $u(s) = u(t) = c$ with $s < t$, monotonicity gives $u(x) \leq c$ for $x < s$, while convexity gives

$$0 \leq \frac{u(s) - u(x)}{s - x} \leq \frac{u(t) - u(s)}{t - s} = 0.$$

Thus $u(x) = c$ for all $x < s$, contradicting $\ell < c$. Uniqueness together with monotonicity gives both stated equivalences. Finally, for any $a < s$, the supporting-slope inequality gives

$$u'(s) \geq \frac{u(s) - u(a)}{s - a} = \frac{c - u(a)}{s - a} > 0. \tag{3.4}$$

□

3.2. **Choice and continuity of the radius.** Since the zero section is compact, choose once and for all

$$c > \max_{z \in Z} F(0_z). \tag{3.5}$$

Lemma 3.1 provides a unique function $\sigma_i : B_i \to \mathbb{R}$ such that

$$U_i(z, \sigma_i(z)) = c. \tag{3.6}$$

For nonzero vectors, $t = \log |\lambda_i|^2$, so

$$\{F < c\} \cap A^{-1}|_{B_i} = \{(z, \lambda_i) : |\lambda_i|^2 < e^{\sigma_i(z)}\}. \tag{3.7}$$

The zero vectors belong to both sides by (3.5). Thus $e^{\sigma_i(z)/2}$ is exactly the fiber radius of this sublevel set.

Continuity does not require an implicit function theorem. Fix z_0, write $s_0 = \sigma_i(z_0)$, and take $\varepsilon > 0$. The strict inequalities

$$U_i(z_0, s_0 - \varepsilon) < c < U_i(z_0, s_0 + \varepsilon)$$

persist for z near z_0 by continuity of U_i. The unique level root must then lie between $s_0 - \varepsilon$ and $s_0 + \varepsilon$. Therefore σ_i is continuous.

4. Positivity of the negative logarithmic radius

Proposition 4.1. *For the continuous radii constructed above, $\psi_i = -\sigma_i$ is psh.*

Proof. Let f be a holomorphic map from a neighborhood of $\overline{\Delta}$ to B_i. The function $\varphi(e^{i\theta}) = \sigma_i(f(e^{i\theta}))$ is continuous on $\partial\Delta$. Its Poisson integral is

$$H(re^{it}) = \frac{1}{2\pi}\int_0^{2\pi} \frac{1 - r^2}{1 - 2r\cos(t-\theta) + r^2}\varphi(e^{i\theta})\, d\theta, \qquad 0 \le r < 1. \tag{4.1}$$

The Poisson kernel is nonnegative, has integral 2π, and concentrates at its boundary point as $r \uparrow 1$. Uniform continuity of φ therefore gives a continuous extension of H to $\overline{\Delta}$, equal to φ on the boundary. The Poisson kernel is harmonic in the interior variable, so H is harmonic and smooth on Δ. In particular,

$$H(0) = \frac{1}{2\pi}\int_0^{2\pi} \sigma_i(f(e^{i\theta}))\, d\theta. \tag{4.2}$$

Writing the disc coordinate as $x+iy$, the function $H_x - iH_y$ is holomorphic because $H_{xx}+H_{yy} = 0$. Since Δ is simply connected, it has a holomorphic primitive g. Adjusting the real constant of integration gives $\operatorname{Re} g = H$. The map

$$\zeta \longmapsto e^{g(\zeta)/2} e_i^*(f(\zeta))$$

is holomorphic on Δ. Hence

$$q(\zeta) = F\big(e^{g(\zeta)/2} e_i^*(f(\zeta))\big) = U_i(f(\zeta), H(\zeta)) \tag{4.3}$$

is subharmonic there. The second expression is continuous on $\overline{\Delta}$, and its boundary value is

$$U_i(f(\zeta), \sigma_i(f(\zeta))) = c.$$

The maximum principle yields $q \le c$. By Lemma 3.1, this is equivalent to

$$H(\zeta) \le \sigma_i(f(\zeta)).$$

At the center, using (4.2) and multiplying by -1, we get

$$-\sigma_i(f(0)) \le \frac{1}{2\pi}\int_0^{2\pi} -\sigma_i(f(e^{i\theta}))\, d\theta.$$

Applying the same argument to every smaller disc proves the submean inequalities for $-\sigma_i \circ f$. Since $-\sigma_i$ is continuous, the holomorphic-disc criterion proves it is psh. □

Remark 4.2. The harmonic conjugate of continuous boundary data need not extend continuously to the boundary. The proof does not assume this: $q = U_i(f, H)$ has the required continuous extension even when g does not. Circle invariance is precisely what removes the imaginary part of g from this boundary argument.

4.1. **Gluing and the sign of the curvature.** On an overlap, suppose $e_j = g_{ij}e_i$, with g_{ij} nowhere-vanishing and holomorphic. Then $e_j^* = g_{ij}^{-1}e_i^*$, and equality of the two descriptions of a vector gives

$$\lambda_i e_i^* = \lambda_j e_j^*, \qquad \lambda_j = g_{ij}\lambda_i.$$

At the level $F = c$, (3.7) therefore gives

$$e^{\sigma_j} = |g_{ij}|^2 e^{\sigma_i}, \quad \sigma_j = \sigma_i + \log|g_{ij}|^2, \quad \psi_j = \psi_i - \log|g_{ij}|^2. \tag{4.4}$$

Consequently

$$\|e_i\|_h^2 = e^{-\psi_i} = e^{\sigma_i} \tag{4.5}$$

defines a positive definite continuous Hermitian metric on A. Its curvature is $dd^c\psi_i \geq 0$. The curvature currents agree because $\log|g_{ij}|^2$ is pluriharmonic locally. This completes the continuous forward implication, with no differentiation of F.

5. Recovering precisely the original regularity

Composition in (3.1) preserves every class in (1.1), since $(z,t) \mapsto e^{t/2}e_i^*(z)$ is real analytic. We now recover the regularity of σ_i from the level equation. The three cases below require different justifications.

5.1. **Classical C^r and real analytic regularity.** Suppose first that U_i is C^1. At each level root, Lemma 3.1 gives

$$\partial_t U_i(z, \sigma_i(z)) > 0. \tag{5.1}$$

For any finite $r \geq 1$, the C^r implicit function theorem applied to $U_i(z,t) - c = 0$ therefore gives $\sigma_i \in C^r$ when $U_i \in C^r$. The local solution agrees with the previously constructed continuous radius by uniqueness. Applying this for every finite r gives the C^∞ conclusion.

When $U_i \in C^\omega$, the *real analytic* implicit function theorem applies. More explicitly, take real base coordinates $x \in \mathbb{R}^{2\dim_{\mathbb{C}} Z}$ near z_0 and put $t_0 = \sigma_i(z_0)$. Complexify the convergent power series of $U_i(x,t) - c$ near (x_0, t_0). The resulting holomorphic function $\widetilde{U}(X,T) - c$ satisfies $\partial_T \widetilde{U}(x_0, t_0) \neq 0$. The holomorphic map

$$(X,T) \longmapsto (X, \widetilde{U}(X,T) - c)$$

has invertible Jacobian there. Its local holomorphic inverse, restricted to the second coordinate 0, supplies a holomorphic function $T = S(X)$ solving the level equation. The complexified series is real on real inputs; uniqueness, or conjugation of this equation, implies that $S(x)$ is real for real x near x_0. Thus its restriction is real analytic and equals σ_i by uniqueness. Formula (4.5) now makes h real analytic.

Remark 5.1. No regular-value selection by Sard's theorem is needed. Every $c > \max_Z F(0_z)$ has the required positive radial derivative. In the C^1 case, this also implies that $F = c$ is a regular level: along $s \mapsto e^s v$ its derivative is $2U_{i,t} > 0$ at such a point.

5.2. **The $C^{0,\alpha}$ case, including Lipschitz regularity.** The implicit function theorem is unavailable at this regularity. Fix z_0 and abbreviate $U = U_i$, $\sigma = \sigma_i$. Choose

$$a < b < \sigma(z_0) < d, \qquad U(z_0, b) > U(z_0, a).$$

Such a, b exist: by continuity there is a $b < \sigma(z_0)$ with $U(z_0, b) > F(0_{z_0})$, and then the limit at $-\infty$ supplies $a < b$ with the strict inequality. Continuity of U and σ permits shrinking the base neighborhood so that

$$b < \sigma(z) < d, \qquad \frac{U(z,b) - U(z,a)}{b-a} \geq m > 0 \tag{5.2}$$

throughout it, with one constant m. Ordered secant slopes of a convex function now imply

$$U(z,t) - U(z,s) \geq m(t-s), \qquad b \leq s \leq t \leq d. \tag{5.3}$$

Assume $\sigma(z) \geq \sigma(w)$. Since both radii lie in (b,d),

$$\begin{aligned} m|\sigma(z) - \sigma(w)| &\leq U(z,\sigma(z)) - U(z,\sigma(w)) \\ &= U(w,\sigma(w)) - U(z,\sigma(w)) \\ &\leq C|z-w|^\alpha. \end{aligned}$$

The last estimate follows from the $C^{0,\alpha}$ bound for U on a compact product of a smaller base neighborhood with $[b,d]$. If the two radii have the opposite order, interchange z, w. Hence

$$|\sigma(z) - \sigma(w)| \leq (C/m)|z-w|^\alpha. \tag{5.4}$$

This proves the same Hölder exponent for the radius, including $\alpha = 1$. Taking its exponential preserves that exponent for h.

5.3. **The $C^{k,\alpha}$ case for $k \geq 1$.** For completeness, one can obtain the Hölder version of the implicit function theorem here directly. The classical theorem first gives $\sigma \in C^k$. On a small convex real coordinate neighborhood, σ is Lipschitz because its first derivatives are bounded. Write x for the real base variables. Differentiating $U(x,\sigma(x)) = c$ once yields

$$D_{x_j}\sigma(x) = -\frac{U_{x_j}(x,\sigma(x))}{U_t(x,\sigma(x))}. \tag{5.5}$$

The denominator has a positive lower bound after shrinking the neighborhood, by (5.1). For $k = 1$, the first derivatives of U are α-Hölder, and composition with the Lipschitz map $x \mapsto (x,\sigma(x))$ preserves this property. Products and reciprocals of bounded Hölder functions with nonzero denominator preserve it as well. Thus $D\sigma$ is α-Hölder.

For $k \geq 2$, differentiating the level equation k times gives, for each multi-index β of length k,

$$U_t(x,\sigma(x))D^\beta\sigma(x) = -P_\beta(x). \tag{5.6}$$

Here P_β is a finite sum of products of derivatives of U of order at most k, evaluated at $(x,\sigma(x))$, and derivatives of σ of order at most $k-1$. This form follows inductively from the chain and product rules: the only term involving a derivative of σ of order k is the displayed $U_t D^\beta\sigma$. Derivatives of σ of order at most $k-1$ are locally Lipschitz since $\sigma \in C^k$. Derivatives of U of order less than k are locally Lipschitz, and those of order k are α-Hölder. Their compositions with $(x,\sigma(x))$ therefore have the same bounds. All factors of P_β are bounded and α-Hölder on the smaller neighborhood, so P_β is α-Hölder. Division by U_t in (5.6) proves that $D^\beta\sigma$ is α-Hölder. This argument works unchanged at $\alpha = 1$.

Together with the continuous construction, these arguments prove Theorem 1.1(i)⇒(ii) for every listed class.

6. From a semipositive metric to an exhaustion

Let h be a metric as in Theorem 1.1(ii), with local weights ψ_i. Define the squared dual norm

$$\rho(v) = \|v\|^2_{h^{-1}}. \tag{6.1}$$

Since $\|e_i^*\|^2_{h^{-1}} = 1/\|e_i\|^2_h = e^{\psi_i}$, its local formula is

$$\rho(z,\lambda_i) = |\lambda_i|^2 e^{\psi_i(z)}. \tag{6.2}$$

Under the frame changes (4.4),

$$|\lambda_j|^2 e^{\psi_j} = |g_{ij}\lambda_i|^2 e^{\psi_i - \log|g_{ij}|^2} = |\lambda_i|^2 e^{\psi_i}.$$

Thus these formulas agree on overlaps.

6.1. **Regularity, including the zero section.** The function $|\lambda_i|^2$ is a polynomial in the real fiber coordinates. Composition of ψ_i with the exponential and multiplication by this polynomial preserve C^r, $C^{k,\alpha}_{\mathrm{loc}}$, and C^ω. Therefore ρ has the stated regularity on the entire bundle chart, including $\lambda_i = 0$. There is no division by λ_i or logarithmic singularity in (6.2). For example, for bounded Hölder functions the elementary estimate

$$|a(x)b(x) - a(y)b(y)| \le \|a\|_\infty |b(x) - b(y)| + \|b\|_\infty |a(x) - a(y)|$$

gives the product bound; applying the product and chain rules gives the higher-order versions. On this noncompact total space, the result is local regularity, not a uniform global Hölder bound.

6.2. **Plurisubharmonicity with no loss at zero.** Away from the zero section,

$$\log \rho = \log |\lambda_i|^2 + \psi_i(z)$$

is psh, because the first summand is pluriharmonic there. The exponential is increasing and convex, so ρ is psh there. Indeed, for a subharmonic function u, its submean inequality and Jensen's inequality give

$$e^{u(0)} \le e^{\frac{1}{2\pi}\int u(e^{it})dt} \le \frac{1}{2\pi}\int e^{u(e^{it})}\, dt.$$

The same argument on holomorphic discs proves the psh assertion.

Here is an explicit local proof across the zero section, avoiding any regularity assumption on second derivatives of ψ_i. For $\varepsilon > 0$ set, on a fixed bundle chart,

$$\rho_\varepsilon(z, \lambda_i) = (|\lambda_i|^2 + \varepsilon) e^{\psi_i(z)}.$$

The function $\log \rho_\varepsilon = \log(|\lambda_i|^2 + \varepsilon) + \psi_i(z)$ is psh, since

$$\frac{\partial^2}{\partial \lambda_i \partial \bar{\lambda}_i} \log(|\lambda_i|^2 + \varepsilon) = \frac{\varepsilon}{(|\lambda_i|^2 + \varepsilon)^2} \ge 0.$$

Thus ρ_ε is psh by convex composition. Moreover $\rho_\varepsilon \to \rho$ locally uniformly, since $\rho_\varepsilon - \rho = \varepsilon e^{\psi_i}$ and ψ_i is continuous. Passing to the limit in the disc submean inequalities shows that ρ is psh throughout the chart. These auxiliary ρ_ε need not agree on overlaps: they are used only to prove the local property of the globally defined ρ.

Equivalently, one may use the removable-singularity theorem for locally bounded psh functions across the zero section. The local argument above verifies directly that the extension has value zero.

6.3. **Compactness of the sublevels.** Choose any smooth reference Hermitian metric k on A^{-1}. Since the fibers are one-dimensional, the ratio

$$a(z) = \frac{\|v\|^2_{h^{-1}}}{\|v\|^2_k}, \qquad 0 \ne v \in A_z^{-1},$$

is independent of the choice of v. It is a positive continuous function on compact Z. Hence there are constants $0 < m \le M < \infty$ such that

$$m\|v\|^2_k \le \rho(v) \le M\|v\|^2_k. \tag{6.3}$$

The closed disk bundle $\{\|v\|^2_k \le R\}$ over a compact base is compact. To check this explicitly, cover Z by finitely many compact sets K_j each contained in a bundle chart. On K_j, the positive coefficient of k has a positive minimum, so the indicated disk bundle lies in $K_j \times \{|\lambda| \le C_j\}$ and is closed there. Each such piece is compact, and their finite union is compact.

For $b \geq 0$, (6.3) places the closed set $\{\rho \leq b\}$ inside the compact disk bundle $\{\|v\|_k^2 \leq b/m\}$. For $b < 0$ the sublevel is empty. Thus ρ is an exhaustion. This proves Theorem 1.1(ii)⇒(i), and completes the theorem.

Remark 6.1 (Geometric normalization). If h was obtained from a circle-averaged exhaustion and a level c, then $\psi_i = -\sigma_i$, so

$$\rho(z, \lambda_i) = |\lambda_i|^2 e^{-\sigma_i(z)}, \qquad \{\rho < 1\} = \{F < c\}.$$

The new exhaustion need not equal F; it is the squared Hermitian gauge of this one complete circular sublevel set.

7. The classical Levi calculations, written out

The preceding proof establishes positivity at continuous regularity. For comparison with the original Proposition 3.3, we verify its C^2 calculation, including the sign and the factor 2.

7.1. **Levi form of the radius.** Assume U and σ are C^2 and omit the chart index. All derivatives of U below are evaluated at $(z, \sigma(z))$. Differentiating $U(z, \sigma(z)) = c$ gives

$$U_{z_j} + U_t \sigma_{z_j} = 0$$

and then

$$0 = U_{z_j \bar{z}_k} + U_{z_j t}\sigma_{\bar{z}_k} + U_{t\bar{z}_k}\sigma_{z_j} + U_{tt}\sigma_{z_j}\sigma_{\bar{z}_k} + U_t \sigma_{z_j \bar{z}_k}. \tag{7.1}$$

For $\xi \in T^{1,0}_z Z$, put $b = \partial\sigma(\xi)$. Since $V(z, w) = U(z, \operatorname{Re} w)$,

$$V_{z_j \bar{w}} = \tfrac{1}{2} U_{z_j t}, \qquad V_{w\bar{z}_k} = \tfrac{1}{2} U_{t\bar{z}_k}, \qquad V_{w\bar{w}} = \tfrac{1}{4} U_{tt}.$$

At the point $(z, w = \sigma(z))$ this gives

$$\begin{aligned} \mathcal{L}_V(\xi, 2b) &= \sum_{j,k} U_{z_j \bar{z}_k} \xi_j \bar{\xi}_k + 2 \operatorname{Re}\left(\sum_j U_{z_j t} \xi_j \bar{b} \right) + U_{tt} |b|^2 \\ &= -U_t \mathcal{L}_\sigma(\xi), \end{aligned}$$

where the last equality is (7.1) contracted with $\xi_j \bar{\xi}_k$. Because V is psh and $U_t > 0$,

$$\mathcal{L}_{-\sigma}(\xi) = \frac{\mathcal{L}_V(\xi, 2\partial\sigma(\xi))}{U_t} \geq 0. \tag{7.2}$$

The plus sign in the vector $(\xi, 2\partial\sigma(\xi))$ is correct.

7.2. **Levi form of the squared dual norm.** For $\rho = |\lambda|^2 e^{\psi}$, direct differentiation yields

$$\begin{aligned} \rho_{z_j \bar{z}_k} &= |\lambda|^2 e^{\psi} (\psi_{z_j \bar{z}_k} + \psi_{z_j} \psi_{\bar{z}_k}), \\ \rho_{z_j \bar{\lambda}} &= \lambda e^{\psi} \psi_{z_j}, \qquad \rho_{\lambda \bar{z}_k} = \bar{\lambda} e^{\psi} \psi_{\bar{z}_k}, \qquad \rho_{\lambda\bar{\lambda}} = e^{\psi}. \end{aligned}$$

Contracting with (ξ, η) and completing the square gives

$$\mathcal{L}_\rho(\xi, \eta) = e^{\psi} \left(|\eta + \lambda \partial\psi(\xi)|^2 + |\lambda|^2 \mathcal{L}_\psi(\xi) \right) \geq 0. \tag{7.3}$$

At $\lambda = 0$ this reduces to $e^{\psi}|\eta|^2$, so it also explicitly checks positivity on the zero section in the analytic and C^2 cases.

8. Audit of the earlier discussion

The continuous radius argument and its Hölder extension are valid. The following qualifications distinguish the established conclusions from stronger statements that do not follow from this proof.

1. **Analytic averaging.** The compactness estimate in Lemma 2.2 supplies uniform Taylor convergence. Smooth differentiation under the integral alone is insufficient. With this addition, the analytic implicit function theorem proves the requested real analytic equivalence.
2. **Low-regularity positivity is not a Hessian computation.** For merely continuous or C^1 data, Proposition 4.1 proves the psh property by harmonic comparison. Curvature is interpreted as a positive current. The second-derivative identity (7.2) is used only when those derivatives exist.
3. **There is no hidden boundary assumption on a harmonic conjugate.** Only H and the expression $U_i(f, H)$ must extend continuously to the closed disc; g need not do so. See Remark 4.2.
4. **Hölder regularity at order zero uses an inverse estimate.** The strictly positive uniform secant slope in (5.2), not an unavailable differentiable implicit function theorem, is what preserves $C^{0,\alpha}$ and Lipschitz regularity. For $k \geq 1$, (5.5)–(5.6) verify the Hölder conclusion, including the endpoint $\alpha = 1$.
5. **An equivalence at stronger regularity does not descend by inclusion alone.** Let $E_{\mathcal{R}}$ denote existence of an exhaustion of class $\mathcal{R}$, and $H_{\mathcal{R}}$ existence of a metric of that class. If $\mathcal{R}_s \subset \mathcal{R}_t$, then
$$E_{\mathcal{R}_s} \Longrightarrow E_{\mathcal{R}_t}, \qquad H_{\mathcal{R}_s} \Longrightarrow H_{\mathcal{R}_t}.$$
Even together with $E_{\mathcal{R}_s} \Longleftrightarrow H_{\mathcal{R}_s}$, these arrows do not imply $E_{\mathcal{R}_t} \Longleftrightarrow H_{\mathcal{R}_t}$: the two weaker existence properties may enlarge independently. The earlier chat's claim that the weaker equivalence follows automatically by inclusion must therefore be removed. The direct proof above establishes the equivalence separately at every class.
6. **The Hölder inclusions themselves need the stated endpoint convention.** For $0 < \alpha < \beta \leq 1$, $C^{k,\beta}_{\mathrm{loc}} \subset C^{k,\alpha}_{\mathrm{loc}}$, since on a compact set of diameter D,
$$|x - y|^{\beta} \leq D^{\beta-\alpha}|x - y|^{\alpha}.$$
Bounded first derivatives on smaller coordinate balls also imply
$$C^{k+1} \subsetneq C^{k,1}_{\mathrm{loc}} \subset C^{k,\alpha}_{\mathrm{loc}} \quad (0 < \alpha < 1).$$
For the strict first inclusion, use $|x|$ when $k = 0$; for $k \geq 1$, take a k-fold antiderivative of $|x|$, whose kth derivative is Lipschitz but not differentiable at zero. The example $|x|^{3/2} \in C^{1,1/2} \setminus C^2$ illustrates a different endpoint issue and does not by itself prove $C^{k+1} \subsetneq C^{k,1}_{\mathrm{loc}}$. For noninteger orders, the usual ordering by $k + \alpha$ is valid; integer differentiability and Lipschitz endpoints must remain distinct.
7. **Existence at every regularity is not asserted.** For each class, the theorem equates two existence questions for a fixed bundle. It does not assert that any of them has a positive answer, nor that C^{∞} semipositivity implies C^{ω} semipositivity. Likewise, nonexistence of a C^2 metric rules out C^2, smooth, and analytic exhaustions, but does not alone rule out C^1 or locally Lipschitz exhaustions. In the particular example of [1, Theorem 1.1], the C^2 obstruction already rules out the analytic class; the lower-regularity threshold requires its own argument.
8. **The hypotheses and signs are essential to the construction.** Compactness of Z supplies one level above the whole zero section and compact norm disk bundles. The object constructed is a metric on A, while the exhaustion lives on A^{-1}. With the convention $\|e_i\|_h^2 =$

$e^{-\psi_i}$, the psh weight is $-\sigma_i$, and the metric coefficient is e^{σ_i}. These signs and the factor 2 in (7.2) agree with the original C^2 proof.

Corollary 8.1 (Replacement for Corollary 3.4)**.** *Let $A \to Z$ be a holomorphic line bundle over a compact complex manifold. For each $r \in \mathbb{Z}_{\geq 0} \cup \{\infty, \omega\}$, $\mathrm{Tot}(A^{-1})$ admits a C^r psh exhaustion if and only if A admits a C^r Hermitian metric with psh local weights. The same equivalence holds with C^r replaced on both sides by $C^{k,\alpha}_{\mathrm{loc}}$, for every $k \in \mathbb{Z}_{\geq 0}$ and $0 < \alpha \leq 1$. Here C^ω means real analytic, and below C^2 semipositivity is understood in the sense of currents.*

Source and scope note. The audit concerns the radius criterion and the regularity discussion in the referenced chat, not the other geometric arguments in the manuscript. The proofs of the averaging, radius, and regularity steps are supplied in this note. Standard background on psh functions, holomorphic pullback, convex composition, and removable singularities can be found in [2, Chapter I, §5]; in that chapter see Definition 5.1 and Theorems 5.6, 5.11, and 5.24.

References

[1] Z. Chen and B. Xiao, *Continuous and Smooth Plurisubharmonic Exhaustions: Threefold Counterexamples and Surface Regularization*, manuscript, October 30, 2026, §3, especially Proposition 3.3 and Corollary 3.4. The version consulted is the local source `10-30-revision/psh-exhaustion.tex`.

[2] J.-P. Demailly, *Complex Analytic and Differential Geometry*, open-content book, Chapter I, §5. Author's PDF.